MarketMind AI

Domain-Specific Marketing AI Agent

Technical Design & Evaluation Document

NationAI Technical Assessment
Domain: Marketing
Synthetic Brand: NovaGlow
Final Local Model: Llama 3.2 1B via Ollama

RAG	LLM	UI	Evaluation
FAISS + MiniLM	Llama 3.2 1B	Streamlit	100% controlled suite

Final submission version

1. Executive Summary

MarketMind AI is a domain-specific marketing assistant built for the synthetic skincare brand NovaGlow. The system combines Retrieval-Augmented Generation (RAG), deterministic fact extraction, local LLM reasoning, structured output validation, and post-generation guardrails. It supports three primary tasks: factual knowledge lookup, marketing campaign generation, and customer-segment analysis.

The final implementation is deliberately local and lightweight. It uses Llama 3.2 1B through Ollama on CPU, SentenceTransformers MiniLM embeddings, FAISS for vector retrieval, LangChain for orchestration, Pydantic for output schemas, and Streamlit for the demo interface. The design emphasizes grounded answers and transparent handling of missing information.

Final evaluation result: All included controlled tests passed: 100% intent accuracy, 100% retrieval hit rate, 100% fact accuracy, 100% hallucination-refusal rate, and 100% structured-output validity. These scores apply only to the included controlled evaluation suite.

2. Assignment Coverage

Requirement	MarketMind AI implementation
LLM reasoning	Local Llama 3.2 1B via Ollama with low-temperature structured generation
RAG	PDF knowledge base -> text chunks -> MiniLM embeddings -> FAISS top-k retrieval
Prompt engineering	System guardrails, task-specific prompts, few-shot campaign example, domain adaptation
Guardrails	Unsupported-fact refusal, entity-aware numeric validation, campaign and segment validators
Structured recommendations / summaries	Pydantic schemas for fact, campaign, and segment responses
Hallucination minimization	Hybrid deterministic + LLM architecture, source tracking, post-generation validation
Demo	Streamlit conversational UI with intent, confidence, sources, and example questions
Evaluation	Intent, retrieval, factual accuracy, hallucination refusal, and structured output tests

3. System Architecture

MarketMind follows a hybrid pipeline rather than sending every request directly to an LLM. The query is first classified, relevant NovaGlow evidence is retrieved, and then the request is routed either through deterministic extraction or constrained LLM reasoning. All LLM outputs are schema-validated and then checked again by domain-specific programmatic guardrails.

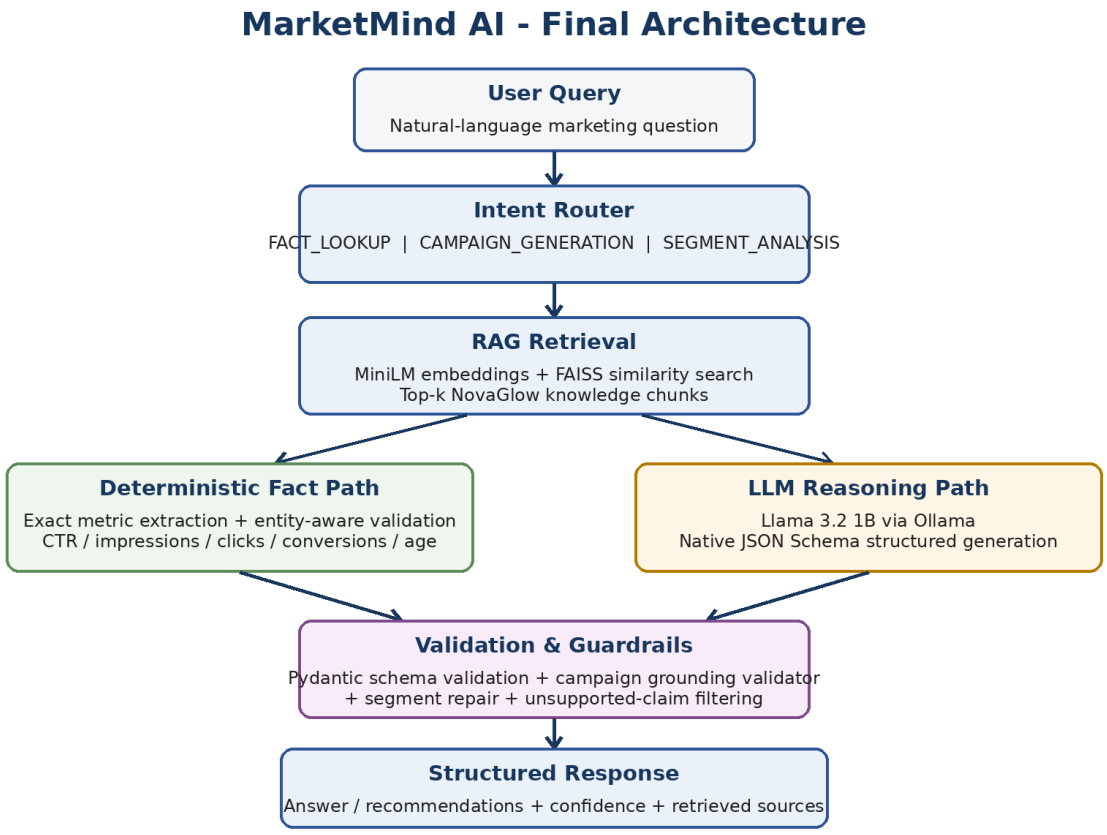


Figure 1. Final MarketMind AI architecture.

Layer	Technology	Role
Interface	Streamlit	Interactive demo, example prompts, source and confidence display
Intent router	Python rules	Routes to FACT_LOOKUP, CAMPAIGN_GENERATION, or SEGMENT_ANALYSIS
Retrieval	LangChain + FAISS	Retrieves top-k relevant knowledge chunks
Embeddings	all-MiniLM-L6-v2	Local semantic embeddings on CPU
Reasoning	Llama 3.2 1B / Ollama	Campaign and segment reasoning plus non-deterministic fact answers
Validation	Pydantic + programmatic validators	Schema enforcement and grounding repair

4. Knowledge Base and RAG Pipeline

The knowledge base is intentionally small, synthetic, and auditable. It contains three PDF documents representing the types of internal marketing material a company could provide.

Document	Purpose	Representative content
brand_guidelines.pdf	Brand and product knowledge	Brand voice, target audience, values, product benefits, product positioning
customer_personas.pdf	Audience knowledge	Young Professional and Skincare Enthusiast demographics, pain points, channels, motivations

Document	Purpose	Representative content
campaign_history.pdf	Historical campaign evidence	Channels, impressions, clicks, CTR, conversions, and observations

4.1 Retrieval Flow

1. Load each PDF with PyPDFLoader and preserve filename/page metadata.
2. Split text into overlapping chunks (approximately 700 characters with 120-character overlap).
3. Embed chunks using sentence-transformers/all-MiniLM-L6-v2 on CPU.
4. Store vectors in a local FAISS index.
5. For each query, retrieve the top four semantically similar chunks.
6. Format retrieved text with source and page labels for downstream reasoning and display.

Why FAISS: The assignment knowledge base is small and local. FAISS provides fast retrieval without external infrastructure, credentials, or usage cost. A managed vector database would be more appropriate for large-scale production deployments.

5. Intent Routing and Hybrid Reasoning

MarketMind uses an explicit intent router before generation. This creates predictable behavior and prevents a generic prompt from handling every task in the same way.

Intent	Typical request	Processing path
FACT_LOOKUP	"What was the CTR of Morning Made Simple?"	RAG + deterministic extraction when supported; otherwise structured LLM lookup
CAMPAIGN_GENERATION	"Create a campaign for HydraBoost Serum..."	RAG + campaign prompt + structured LLM + campaign grounding validator
SEGMENT_ANALYSIS	"Analyze the Young Professional persona."	RAG + segment prompt + structured LLM + segment repair/validation

5.1 Deterministic Fact Extraction

A key design decision is to avoid relying on the LLM for explicit numerical facts when those facts can be extracted deterministically. CTR, impressions, clicks, conversions, and age ranges are matched directly from retrieved text. Entity words from the query must also appear near the candidate metric.

Hallucination protection example: For a question about a Facebook conversion rate, MarketMind does not reuse a conversion value from another campaign simply because a conversion number was retrieved. If the requested entity is not located near the metric, the deterministic path returns an unsupported result and the agent refuses to invent an answer.

6. Prompt Engineering and Structured Outputs

Each intent has a dedicated prompt instead of one large general prompt. The shared guardrail block defines NovaGlow knowledge as the primary truth source and explicitly prohibits invented company metrics, prices, product features, medical claims, and cross-product fact transfer.

6.1 Campaign Prompt

- Defines three explicit idea classifications: knowledge_based, reasonable_inference, and general_recommendation.
- Includes a few-shot example using the historical observation that short educational videos and simple morning-routine content performed well.
- Forces general recommendations to carry no evidence and prevents unsupported tactics from being presented as historical company knowledge.
- Separates brand-level traits such as "science-driven" from stronger claims such as "clinically proven" or "dermatologist approved."

6.2 Native JSON Schema + Pydantic

The final version uses Ollama-native JSON-schema generation through LangChain structured output, followed by Pydantic validation. If the first response fails schema parsing, the system retries once with stricter instructions. This was necessary because a small local model can occasionally omit or mis-type required fields.

```
ChatOllama(model="llama3.2:1b", temperature=0, num_gpu=0, num_ctx=4096, num_predict=768)

llm.with_structured_output(schema, method="json_schema", include_raw=True)
```

7. Guardrails and Hallucination Reduction

Hallucination control is implemented at several layers rather than being delegated to prompt wording alone.

Guardrail layer	Behavior
Prompt-level	Prohibits invented metrics, product facts, prices, health/clinical claims, and fact transfer
Deterministic fact path	Extracts explicit numbers and validates entity proximity
Schema validation	Rejects malformed structured responses
Campaign validator	Downgrades unsupported tactics, removes fake evidence, blocks inflated product claims, rebuilds reasoning
Segment validator	Repairs incomplete labels, restores supported persona fields, removes duplication
Source control	Displayed source labels come from retrieval metadata, not from model-generated filenames
Confidence control	High for directly supported facts; Medium where inference/general recommendations are involved; Low for missing evidence

Explicit refusal behavior: When the knowledge base does not support a requested fact, MarketMind returns a refusal such as: "The available knowledge base does not contain this information." This is preferred over guessing or transferring a nearby fact.

8. Evaluation Strategy and Results

The evaluation suite was designed to test separate failure modes instead of reporting one aggregate number only. The final suite contains intent-routing checks, retrieval checks, factual tests, hallucination-refusal tests, and structured-output checks.

Metric	Final score	What it tests
Intent Accuracy	100.0%	Correct routing between fact, campaign, and segment tasks
Retrieval Hit Rate	100.0%	Whether the expected NovaGlow source appears in retrieved top-k chunks
Fact Accuracy	100.0%	Correct explicit values such as 4.4% CTR, 420,000 impressions, and persona age
Hallucination Refusal Rate	100.0%	Refusal of unsupported Facebook conversion, product price, and clinical-trial claims
Structured Output Validity	100.0%	Pydantic-valid output for fact, campaign, and segment paths
Overall	100.0%	Aggregate across the included controlled evaluation suite

Interpretation: The 100% result is not a claim of universal model accuracy. It means all cases in the included controlled evaluation suite passed. Broader real-world testing would require a larger and more diverse benchmark.

9. Evaluation-Driven Development

The strongest engineering improvement came from treating evaluation failures as design signals rather than simply changing prompts.

Stage	Observed issue	Engineering response
Initial RAG + LLM	Retrieval hit rate was high, but the LLM over-refused explicit numeric questions.	Added deterministic extraction for supported metrics.
After deterministic extraction	Fact accuracy improved, but a Facebook conversion question incorrectly reused another campaign value.	Added entity-aware validation requiring query identifiers near the metric.
Small-model campaign output	The 1B model occasionally promoted influencer/contest ideas as if evidence supported them.	Added campaign grounding validator to classify unsupported tactics as general recommendations and remove evidence.
Small-model segment output	The model sometimes returned labels such as "age" / "occupation" instead of values or duplicated pain points.	Added segment post-validation to repair supported fields from retrieved context and deduplicate output.

This progression demonstrates why the final system is hybrid: deterministic components are used where exactness is more important than generative flexibility, while the LLM is reserved for tasks that benefit from synthesis and creative reasoning.

10. Design Decisions and Trade-offs

10.1 Why Llama 3.2 1B?

The original local configuration used a larger Llama 3.2 model, but the available machine produced memory-allocation failures during campaign and segment generation. The final version therefore uses Llama 3.2 1B. This improves demo reliability and keeps the system fully local, but the smaller model requires stronger prompt constraints and more programmatic post-validation.

Decision	Benefit	Trade-off
Local Llama 3.2 1B	No API cost or external dependency; fits available hardware	Lower reasoning robustness than a larger model
FAISS	Simple, fast, offline, easy to reproduce	Not designed for multi-tenant distributed scale
MiniLM embeddings	Small and fast on CPU	Lower semantic capacity than larger embedding models
Rule-based intent router	Deterministic and easy to test	Needs maintenance as task vocabulary expands
Programmatic validators	Reliable hallucination reduction	Some rules are domain-specific and must evolve with the KB
Synthetic KB	Safe, auditable, reproducible	Does not represent all ambiguity/noise in enterprise data

11. Scaling Toward Production

A production version would preserve the same logical stages while replacing local components where scale or governance requires it.

- Replace FAISS with a managed or distributed vector database and add metadata filters for product, persona, date, market, and campaign.
- Use a larger production LLM or managed inference endpoint, while retaining schema validation and deterministic fact paths.
- Add reranking and retrieval-quality monitoring for larger knowledge bases.
- Introduce document versioning, ingestion pipelines, permissions, audit logs, and source-level access control.
- Expand evaluation into a versioned benchmark with adversarial prompts, regression tests, and human quality review.
- Add observability for latency, retrieval quality, refusals, invalid outputs, and grounding-validator interventions.
- Add conversation memory only for non-sensitive preferences, keeping factual company knowledge grounded in the current retrieval layer.

12. Demo Flow

The Streamlit interface exposes the main capabilities through example buttons and free-form chat. A concise interview demo can use the following sequence:

Step	Prompt	Expected behavior
1	What was the CTR of the Morning Made Simple campaign?	FACT_LOOKUP -> 4.4%, High confidence, campaign-history source

Step	Prompt	Expected behavior
2	Analyze the Young Professional persona.	SEGMENT_ANALYSIS -> grounded age, occupations, pain points, channels and motivations
3	Create a marketing campaign for HydraBoost Serum targeting young professionals.	CAMPAIGN_GENERATION -> grounded/inferred/general idea classification with Medium confidence
4	What was NovaGlow's Facebook campaign conversion rate?	FACT_LOOKUP -> Low-confidence refusal because the fact is not in the KB

13. Current Limitations

- The knowledge base is small and synthetic; retrieval behavior should be re-evaluated on real enterprise documents.
- Rule-based routing and validators are tailored to the current marketing scope and vocabulary.
- The local 1B model can still produce shallow recommendations; validators improve safety but cannot create missing business knowledge.
- The evaluation suite is controlled and intentionally compact; 100% on this suite does not imply universal correctness.
- The current demo is single-user and local; authentication, authorization, persistence, and production monitoring are out of scope.

14. Conclusion

MarketMind AI satisfies the assignment goal of a domain-specific marketing agent that combines LLM reasoning, RAG, prompt engineering, structured outputs, evaluation, and hallucination guardrails. The final design deliberately uses a hybrid architecture: exact facts are handled deterministically when possible, while Llama 3.2 1B is used for synthesis and recommendations under schema constraints. Post-generation campaign and segment validators provide an additional safety layer that makes the small local model more dependable for demonstration and controlled evaluation.

Final status: The final controlled evaluation suite passes all included tests. The implementation is ready for demo and submission, subject to the limitations described above.

Appendix A. Final Technology Stack

Component	Final choice
LLM	Llama 3.2 1B
LLM runtime	Ollama
LLM execution	CPU only (num_gpu=0)
Context window setting	4096
Generation limit	768 tokens
Embeddings	sentence-transformers/all-MiniLM-L6-v2
Vector store	FAISS

Component	Final choice
Orchestration	LangChain
PDF loader	PyPDFLoader
Structured validation	Pydantic + Ollama JSON Schema
UI	Streamlit
Evaluation output	tests/evaluation_report.json

Appendix B. Repository Structure

```
marketmind-ai/
├── data/
│   ├── brand_guidelines.pdf
│   ├── campaign_history.pdf
│   └── customer_personas.pdf
├── src/
│   ├── agent.py
│   ├── evaluation.py
│   ├── rag.py
│   └── schemas.py
├── tests/
│   ├── evaluation_questions.json
│   └── evaluation_report.json
├── app.py
├── README.md
├── requirements.txt
└── .gitignore
```